Expense Tracker

The same app, built in Python and C++

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MSCS632 | Advanced Programming Languages

Python **C++**

What can the app do?

A simple way to record and review everyday spending

01 Add an expense

Date, amount, category, and description

02 Find an expense

View the list or search by a word

03 Filter the list

Choose a date range or a category

04 See totals

Spending by category and overall

A simple example

Start with one everyday purchase

Lunch, campus

$12.34

Food
October 2, 2026

1. Record a lunch purchase
2. Find expenses in Food
3. See how much was spent

How are expenses saved?

Both versions use the same file format

Enter an expense

The app checks the date, amount, and text.

Save a CSV file

A plain text file with rows and columns.

Open it later

Either version can read the saved expenses.

Store $12.34 as 1,234 cents to keep totals exact.

Python: see it working

View the saved expenses and calculate totals

Python output | sample data

```
Category | Total (USD)
Food | $21.09
Supplies | $19.99
Transport | $20.00
Overall | $61.08
```

- 5 sample expenses
- Food total: $21.09
- Overall total: $61.08

Recorded program output. Live demo commands are in the speaker notes.

C++: the same results

A separate program reads the same sample file

C++ output | sample data

```
Category | Total (USD)
Food | $21.09
Supplies | $19.99
Transport | $20.00
Overall | $61.08
```

- Same saved expenses
- Same category totals
- Same overall $61.08

Recorded program output. Live demo commands are in the speaker notes.

One clear code difference

Both records have the same fields

Python

```
class Expense(TypedDict):
    id: int
    date: date
    amount_cents: int
    category: str
    description: str
```

Python: hints describe the fields.

C++

```
struct Expense {
    std::int32_t id;
    std::string date;
    std::int64_t amount_cents;
    std::string category;
    std::string description;
};
```

C++: field types are checked during the build.

What felt different?

Our choices while building this project

In our project	Python	C++
Starting the app	Run using Python	Compile, then run
Expense records	Named fields with type hints	Fields with fixed types
Dates and CSV files	Built-in tools helped	More code written by us
Organizing the code	Separate functions	A Tracker class groups operations

How did we check it?

62 tests passed in our recorded verification run

62

tests passed

- Totals, searches, and date filters
- Invalid dates and amounts
- Saved files used by both versions

What we learned

Same features. Different ways to write the code.

- Python made quick changes convenient.
- C++ made field types and structure explicit.
- Both versions use the same files and give the same results.

Source code, sample data, tests, and run instructions:

https://github.com/psahay101/cross-language-expense-tracker